SHOHINUR PERVEZ SHOHAN

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RESEARCH INTERESTS

I am interested in making machine learning models both efficient and reliable. My current focus is measuring the real energy use of compressed neural networks (quantized and pruned models) on actual hardware, checking how trustworthy common energy-measurement tools are, and finding out whether compressed models still give accurate and well-calibrated predictions after deployment.

EDUCATION

B.Sc. (Engineering) in Computer Science and Engineering *Mar 2018 – Feb 2025*

Rangamati Science and Technology University (RMSTU), Rangamati, Bangladesh

- CGPA: 3.21 / 4.00. My results improved every year: 2.82 (1st year), 3.15 (2nd), 3.32 (3rd), 3.61 (final year)
- Undergraduate thesis on land-cover change in Rangamati from 1993 to 2023 using Landsat images and machine learning, graded A+. Supervisor: Dhiman Sarma, Associate Professor, Dept. of CSE

RESEARCH EXPERIENCE

Energy Measurement of Compressed Vision Models (Phase 1), Independent Research *2026*

- Built my own measurement setup on my laptop (Intel i5-13450HX CPU, RTX 3050 GPU) using Intel RAPL for the CPU and NVIDIA NVML for the GPU. Tested ResNet-18, MobileNetV3-Small and EfficientNet-B0 on CIFAR-10, with INT8 quantization and 30%, 50% and 70% structured pruning
- Main result, from 6 sessions and 1,440 measurement windows: the reduction in MACs (multiply-accumulate operations) overestimated the actual CPU energy saving by about 2.28 times on average (95% CI: 2.26 to 2.30), at batch size 1
- Found two measurement problems in my own setup and corrected them: NVML's built-in energy counter was over-reporting GPU power by about 30%, so I switched to sampled power readings (checked to within 0.02 W); and two RAPL power domains were being added together by mistake, so I recalculated CPU energy from the raw data
- Also checked accuracy: only ResNet-18 with INT8 quantization kept at least 99% of the original accuracy. All nine pruned models without fine-tuning failed, and some dropped close to random guessing. This is the main reason I want to study reliability next
- Fixed my analysis plan in a git tag before collecting data and kept a log of every deviation from it. Plan archived on Zenodo (10.5281/zenodo.23357783). Paper in preparation for *Sustainable Computing: Informatics and Systems*

Literature Review on Energy Measurement of Compressed Deep Learning Models *2026*

- Reviewed more than 50 studies on how energy is measured for deep learning models, the tools used (software estimators, hardware counters, external power meters), and why lower FLOPs does not always mean lower energy. Currently under review (see Publications)

Land-Cover Mapping of the Chittagong Hill Tracts, extended from B.Sc. thesis *2024 – 2026*

- Compared SVM (78.04%), Random Forest (77.10%) and CART (72.90%) using spatially blocked validation, so that test areas were kept separate from training areas and the accuracy was not overestimated
- Used area-adjusted accuracy assessment and tested how the results change under different definitions of forest. Found that total forest area grew by 4.4% (+20,693 ha) between 1993 and 2023, while dense forest fell by 8,235 ha
- Rewrote the thesis into a full conference paper as first author (ICCIT 2026, under review)

PUBLICATIONS AND PREPRINTS

- S. P. Shohan, "Energy Measurement of Compressed Deep Learning Models: A Literature Review," *Artificial Intelligence Theory and Applications (AITA)*, under review, submitted Oct 2026. Preprint: doi.org/10.5281/zenodo.22842954

- S. P. Shohan et al., "Multi-Decadal Land-Cover Mapping in the Chittagong Hill Tracts Using Spatially Blocked Classification, Area-Adjusted Accuracy Assessment, and Forest-Definition Sensitivity Analysis," *International Conference on Computer and Information Technology (ICCIT 2026)*, under review. First author

PROPOSED MASTER'S RESEARCH

I want to study whether quantization and pruning keep a model's accuracy, calibration and robustness under distribution shift, and to report these results together with energy measured on real hardware instead of estimated from MAC counts. My Phase 1 work gives me a tested measurement setup to start from.

TECHNICAL SKILLS

- **Programming:** Python, R, Java, SQL; Linux, Git
- **Machine learning:** PyTorch and torchvision, INT8 post-training quantization, structured pruning (torch-pruning), scikit-learn, OpenCV
- **Measurement and statistics:** Intel RAPL, NVIDIA NVML; mixed-effects models (lme4 in R), confidence intervals, sensitivity analysis
- **Geospatial and data tools:** Google Earth Engine, QGIS, Power BI, Excel

WORK EXPERIENCE

IT Support Assistant, Pro Better Life Bangladesh, Rangamati *Jun 2025 – present*

- Fix hardware, software and network problems, look after the office network, and train colleagues in basic PC troubleshooting

Data Analytics Intern (remote), Codveda Technologies, India *Apr – May 2025*

- Analysed retail sales data and ran hypothesis tests on sales uplift

Data Analytics Intern (remote), Intern Intelligence, Azerbaijan *Mar – Apr 2025*

- Did customer segmentation on the iFood marketing dataset

LANGUAGES AND TEST SCORES

- English: IELTS Academic 6.5 overall (Listening 6.0, Reading 6.5, Writing 6.5, Speaking 6.0), September 2025
- Japanese: NAT-TEST 5Q (equivalent to JLPT N5), August 2026. Bangla: native.

CERTIFICATIONS

Google Data Analytics (Coursera); Code in Place, Python Programming (Stanford University); Data Analyst with Python (DataCamp); Job Simulations by Accenture, Deloitte and Quantium (Forage, 2025)

REFEREE

Dhiman Sarma, Associate Professor, Department of CSE, RMSTU (my B.Sc. thesis supervisor).

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